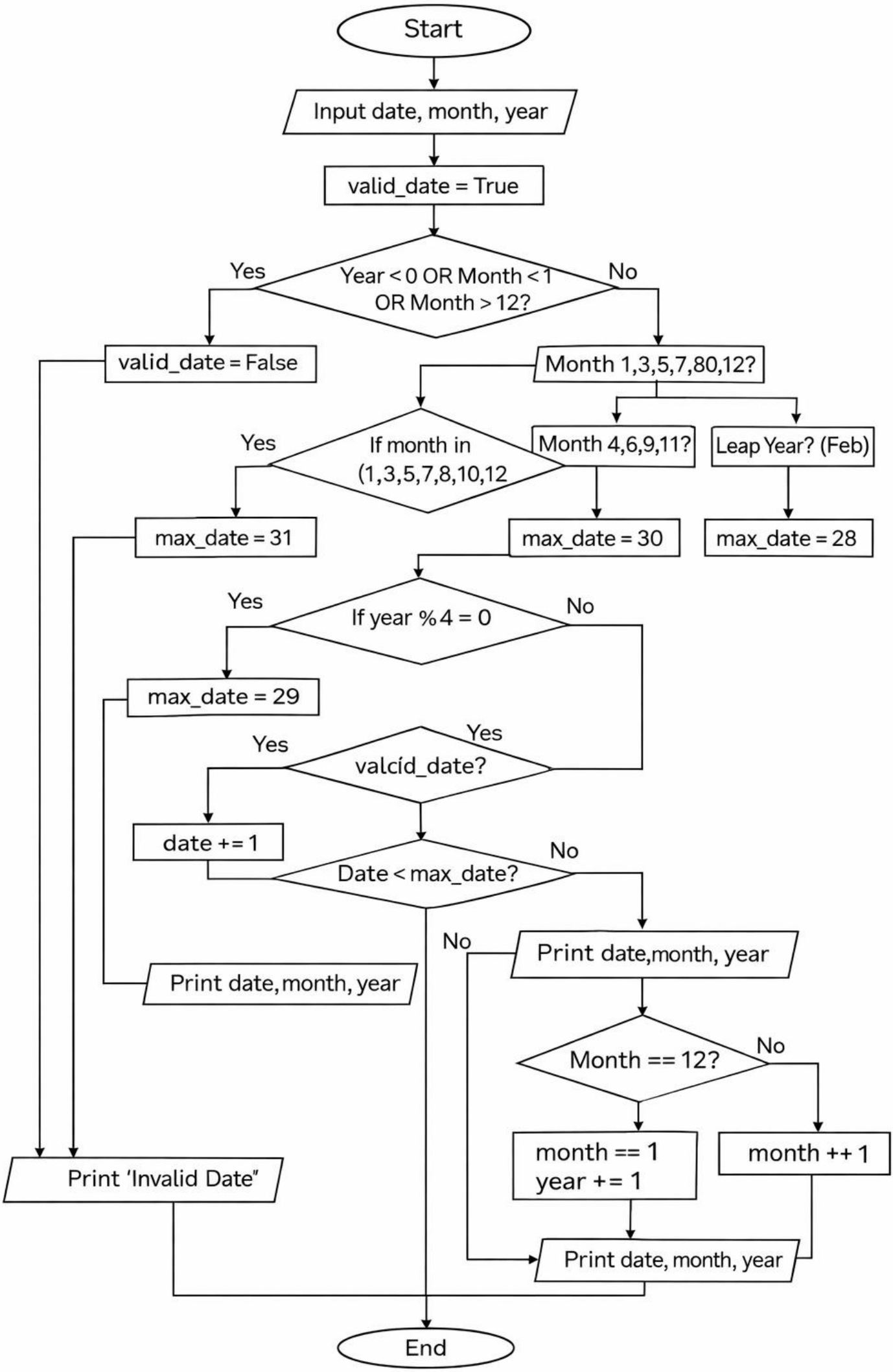
Experiment 6

6.1.1 Incremented Date

Algorithm :

```
Step 1: Start
Step 2: Input date, month, year
Step 3: valid_date = True.
Step 4: If (year <= 0 or month > 12 or
month < 1)
                    then valid_date = False
Step 5: If month is 1,3,5,7,8,10,12
                    then max_date = 31
               else if month is 4,6,9,11
                    then max_date = 30
          else
               if year % 4 == 0
                         max_date = 29
               else
                         max_date = 28
Step 6: If (date > max_date or date < 1)
                    then valid_date = False
Step 7: If valid_date = True
               if date < max_date
                          date = date + 1
               else
                         date = 1
                          if month == 12
                         month = 1
                         year = year + 1
                         else
                    month = month + 1
Step 8 :Print next date (date-month-year)
Step 9: Else
                    Print "Invalid Date".
Step 10: Stop
```

Flowchart :


Start
Input date, month, year
valid_date = True
Year < 0 OR Month < 1
OR Month > 12?
Yes
No
valid_date = False
Month 1,3,5,7,80,12?
If month in
(1,3,5,7,8,10,12
Yes
Month 4,6,9,11?
Leap Year? (Feb)
max_date = 31
max_date = 30
max_date = 28
If year % 4 = 0
Yes
No
max_date = 29
valcid_date?
Yes
Yes
date += 1
Date < max_date?
No
Print date,month, year
No
Print date,month, year
Month == 12?
No
month == 1
year += 1
month ++ 1
Print 'Invalid Date"
Print date, month, year
End

Execution :

CODETANTRA Home amaan.bhurani.batch2025@sitnagpur.siu.edu.in Support Logout

6.1.1. Incremented Date

22:41

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. **Valid Month**: 1 to 12
2. **Valid Day**: Depends on the month and year
   - January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
   - April (4), June (6), September (9), November (11): 1 to 30 days
   - February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
3. **Valid Year**: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Sample Test Cases +

Explorer

nextDate.py

Submit

```python
date = int(input())
month = int(input())
year = int(input())

valid_date = True

if (year <= 0 or month > 12 or month < 1):
    valid_date = False
```

Debugger

Average time	Maximum time
0.007 s	0.009 s
7.30 ms	9.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

- Test case 1 9 ms
- Test case 2 8 ms
- Test case 3 6 ms
- Test case 4 8 ms

Terminal Test cases

< Prev Reset Submit Next >